Weekly Progress: Embedding Results

On Development of an on-Premise AI Product Recommendation Model Using Open-Source LLM

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Project Description

Project Title

Development of an on-Premise AI Product Recommendation Model Using Open-Source LLM

Company: Narosu Co. Ltd.

Project Description

- The e-commerce market is becoming increasingly competitive, requiring advanced recommendation systems.
- Large marketplaces use sophisticated AI models, but independent online store operators lack such solutions.
- This project aims to develop an **AI-powered product recommendation system** tailored for self-managed online stores.
- By analyzing customer preferences, the system will suggest products that align with their interests.

Project Objective

- Develop an AI-based search and recommendation system that analyzes user input sentences and recommends the **TOP 5** most relevant products based on semantic similarity.
- Utilize Natural Language Processing (NLP) and vector search technology to provide a highly refined search experience.
- Deploy in an on-premise environment to ensure data privacy and full control over the recommendation process.

Company Requirements

Key Technology Stack

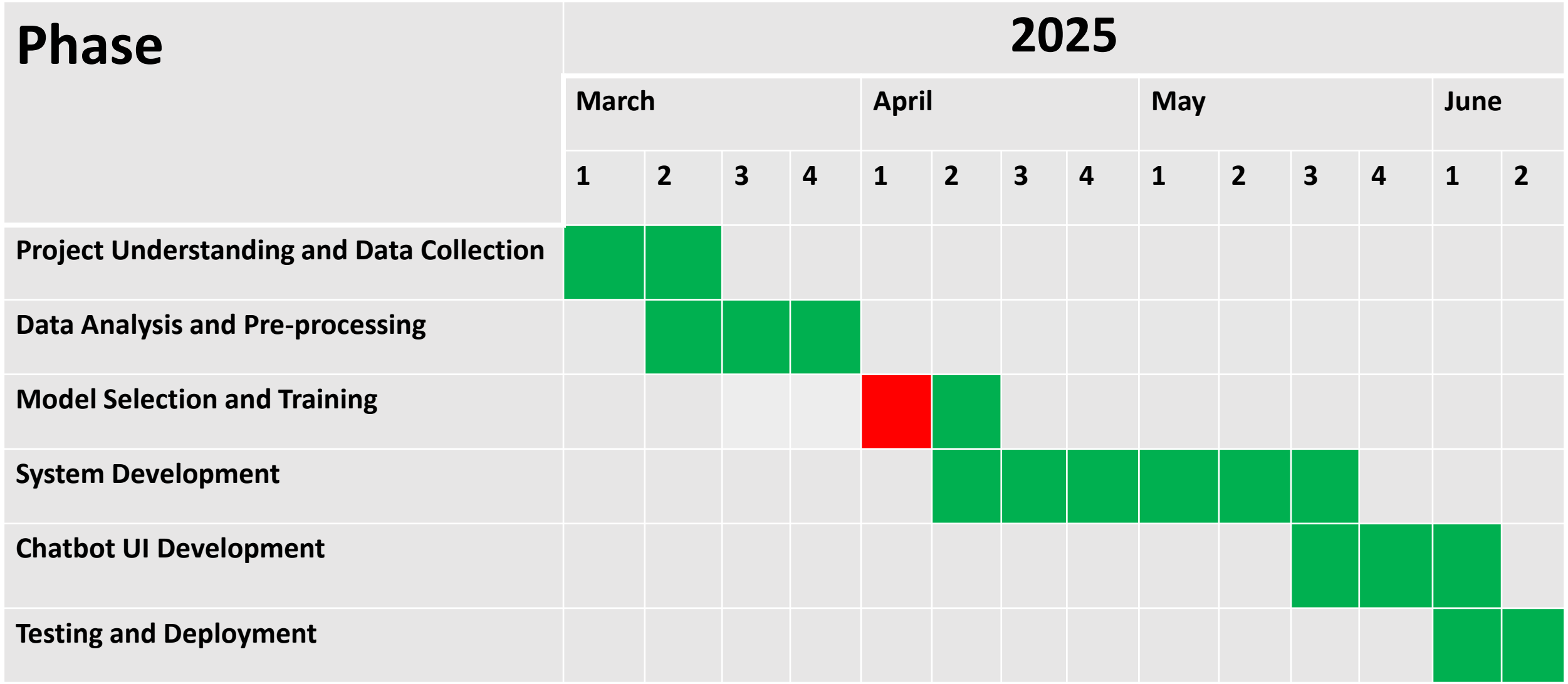
Component	Technology Used	Description
LLM (Large Language Model)	Llama 3.1 8B	Understands and generates natural language based on context
LangChain (LLM Framework)	LangChain	Enhances AI responses by connecting LLM to external data sources
Embedding Model	OpenAI text-embedding-ada-002	Generates text embeddings based on semantic similarity
Vector DB	Flexible (Any compatible DB)	Stores high-dimensional vector data
Vector Search Library	FAISS	Optimized large-scale vector data search
RAG (Retrieval-Augmented Generation)	RAG-based search + LLM	Retrieves relevant documents in real time to generate accurate answers
API Architecture	REST API (FastAPI, Flask, or Django)	Builds an API for client-server data communication
GUI (Graphical User Interface)	Chatbot	Applies a chatbot template for a user-friendly and intuitive interface

- OpenAI text-embedding-ada-002 is a paid model

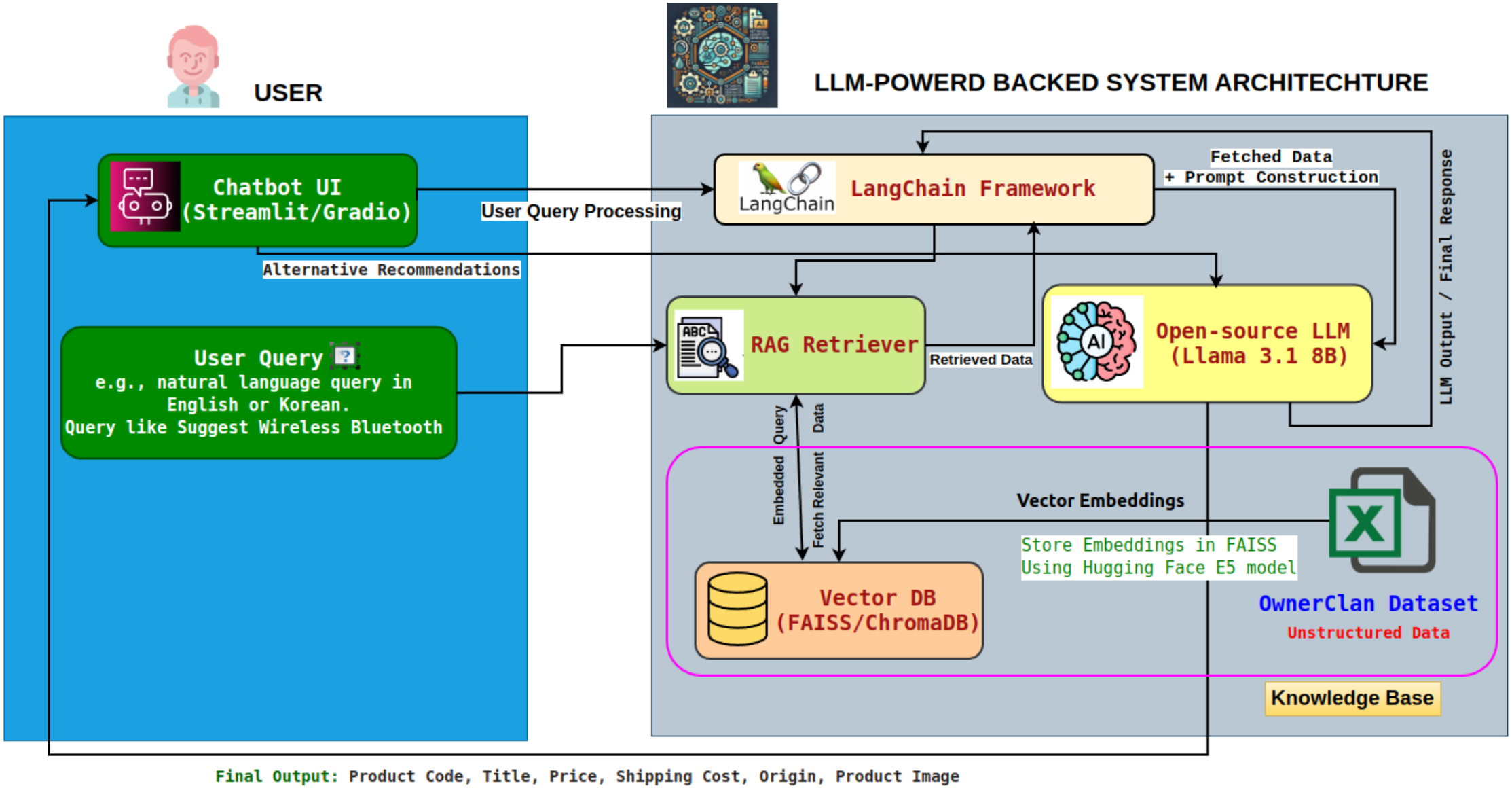
Try with free models such as:

- Sentence Transformers (**`all-MiniLM-L6-v2`**)
- Hugging Face E5-small Transformers (**`intfloat/e5-small-v2`**)
- E5-multilingual-large: **intfloat/multilingual-e5-large**
- LaBSE: **sentence-transformers/LaBSE**

Project Schedule (Gantt Chart)



System Configuration (AI Model) Diagram



Communication with company

Questions:

- Looking at the core technology stack you sent me, it uses FAISS as a vector search library, but is it okay to use something else?

Contents of the answer

- In addition to FAISS (vector search library), you may use vectorDB or other vector search libraries.

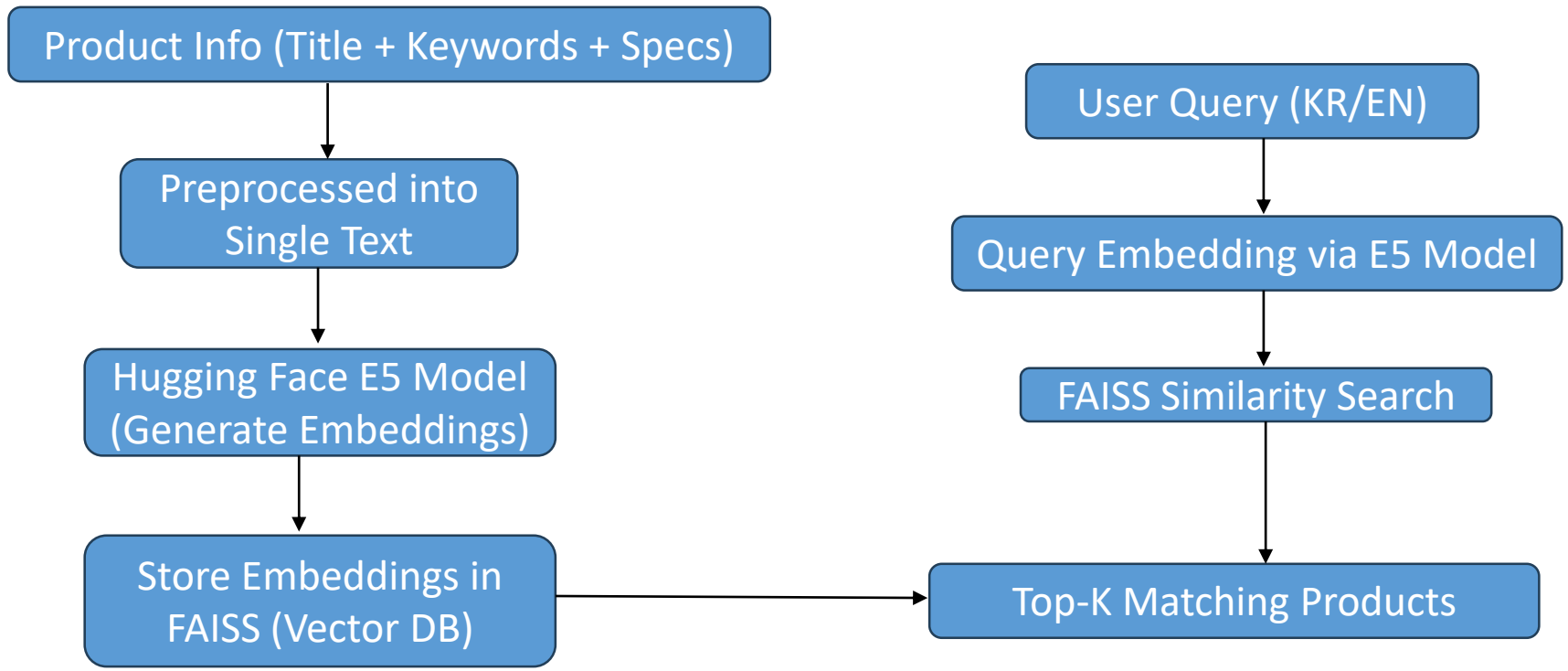
Embeddings in our system (OwnerClan Dataset)

🔍 What are Embeddings?

- Embeddings are **numerical representations** of text data.
- Similar meanings produce **similar vector representations**.
- Example: "블루투스 이어폰" and "Bluetooth earphone" → close in vector space.

🎁 How Embeddings Work

- **Input**: Product title + keywords + specs → combined text
- **Model**: Free model generates vector embeddings
- **Storage**: Vectors stored in FAISS (vector DB)
- **Search**: User query → embedded → matched with product vectors



Example of Embeddings Output in our system

Original Query (KR): 게이밍 블루투스 헤드셋

Translated Query (EN): Gaming Bluetooth Headset

all-MiniLM-L6-v2 and intfloat/e5-small-v2
Result on Korean Dataset

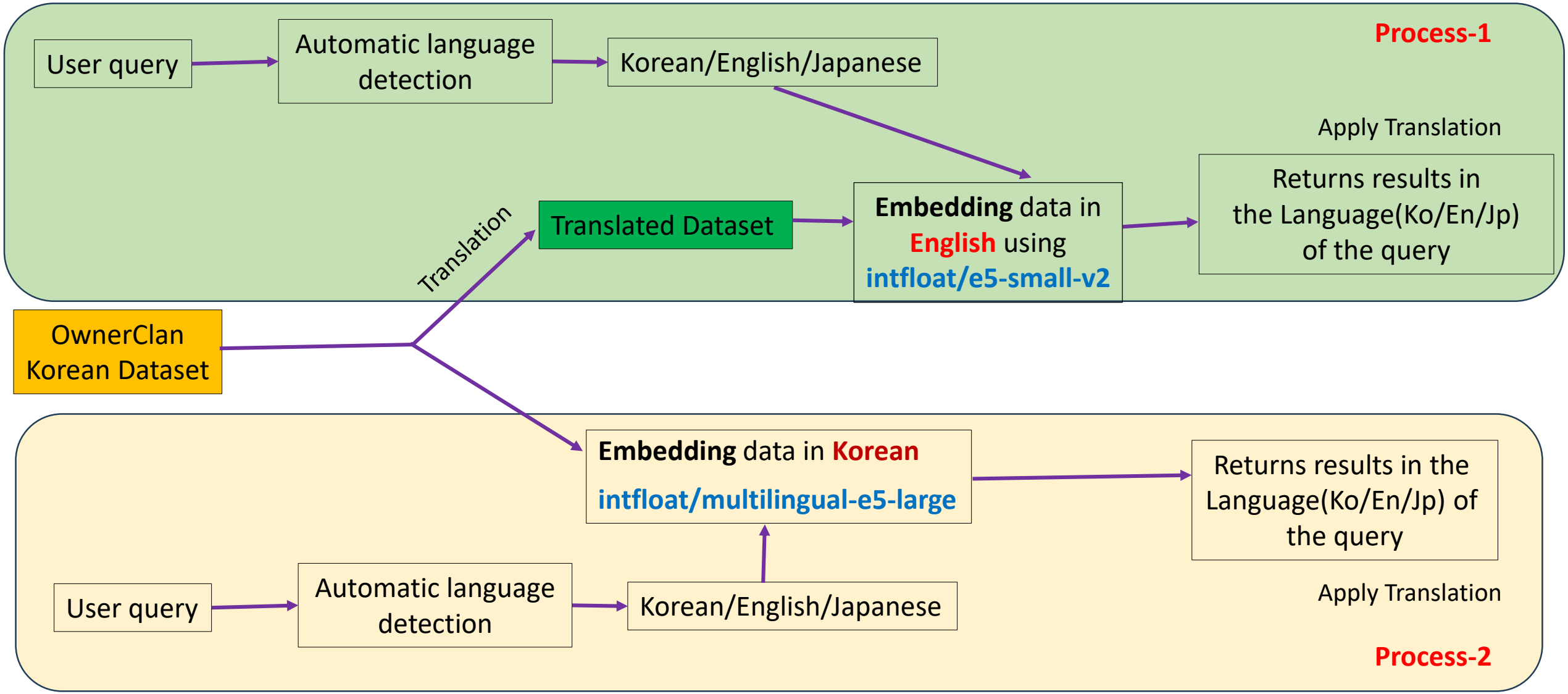
원본상품명 Product Name (EN)	키워드 Keywords (EN)	카테고리명 Category (EN)
6953 맥스틸 올인원 게이밍 헤드셋 GM-GHC300 헤드셋, Max Steel All-in-One Gaming Headset GM-GHC300 Headset	블루투스헤드셋,무선헤드셋,유선헤드셋,게이밍헤드셋 Bluetooth headset, wireless headset	led헤드셋,올인원헤드셋,삼... 디지털/가전>멀티미디어장비>PC헤드셋 Digital/Home Appliances> Multimedia Equipment

```
Original Query (KR): 갤럭시 스마트폰
Translated Query (EN): Galaxy smartphone

                원본상품명                                키워드  ...                                Keywords (EN)                                Catego
ry (EN)
6953  맥스틸 올인원 게이밍 헤드셋 GM-GHC300  헤드셋,블루투스헤드셋,무선헤드셋,유선헤드셋,게이밍헤드셋,led헤드셋,올인원헤드셋,삼...  ...  Headset, Bluetooth headset, wireless headset, ...  Di
gital/Home Appliances> Multimedia Equipment>...
```

Query output from this embedding sometimes provides incorrect results.

Embedding Process and Model Selection



Embedding with Process 1

Original Query: 패딩 크로스백

Translated to English: Padded cross bag (Lang Detected: ko)

	title	keywords	category	specs	similarity_score
2	여자 패딩 크로스 바디 백 인기있는 겨울 패션 가방 숄더백	새로운 겨울 가방, 겨울 가방, 패딩 가방, 겨울 크	패션 액세서리> 여자 가방> 크	유형 : 001202001Q 주제 : 나일	0.884442329406738
3	인기있는 겨울 가방 여성용 패딩 토트 가방 매일 크로스 바디 백	여자 토트 가방, 여자 핸드백, 여성 20 큰 가방,	패션 액세서리> 여자 가방> 크	유형 : 001202006Q 주제 : 나일	0.883921921253204
4	가벼운 겨울 패딩 크로스 바디 백 새로운 도착 가방 여성 어깨 가방	미니 크로스 백, 여자 숄더백, 데일리 가방, 여자	패션 액세서리> 여자 가방> 숄	유형 : 001202007Q 주제 : 나일	0.878635048866272
5	패션 패딩 보스턴 가방 큰 크기 어깨 가방 새로운 더플 백	체육관 가방, 요즘 트렌디 한 가방, 여자 핸드백,	패션 액세서리> 여행 가방 / 소품	유형 : 001202008Q 주제 : PU	0.870376706123352
6	인기있는 크로스 바디 백 매일 새로운 가방 여자 남자 숄더백	미니 크로스 백, 여자 숄더백, 데일리 가방, 여자	패션 액세서리> 여자 가방> 크	유형 : 001202005Q 주제 : 옥스	0.864968836307526

Embedding with Process 2

- Original Query: 패딩 크로스백
- Translated to English: Padded cross bag

	원본상품명	키워드	카테고리명	similarity_score
2	여성 패딩 크로스백 인기 겨울 패션 가방 숄더백	신상겨울가방,겨울가방,패딩가방,가	패션잡화>여성가방>크로스백	0.8760142326355
3	가벼운 겨울 패딩 크로스백 신상 가방 여성 숄더백	미니크로스백,여성숄더백,데일리백	패션잡화>여성가방>숄더백	0.8665043711662
4	인기 겨울 가방 여자 패딩 토트백 데일리 크로스백	여성토트백,여성핸드백,여성20대가	패션잡화>여성가방>크로스백	0.8530597686768
5	패션 핸드백 여성 대학생 크로스백 인기 숄더백 가방	여성토트백,여성핸드백,여성20대가	패션잡화>여성가방>크로스백	0.8529251813889
6	인기 크로스백 데일리 신상 가방 여자 남자 숄더백	미니크로스백,여성숄더백,데일리백	패션잡화>여성가방>크로스백	0.8458325266838

- Original Query: 화이트보드
- Translated to English: White board

	원본상품명	키워드	카테고리명	similarity_score
2	두문 화이트보드(탁상용) 700x500-900x600	부품,소모품,자재,파츠	생활/건강>문구/사무용품>문구용품>화	0.8327831029892
3	두문 법랑화이트보드 700x500 900x600	부품,소모품,자재,파츠	생활/건강>문구/사무용품>문구용품>화	0.8174697756767
4	두문 크리너 화이트보드용(120ml) 12EA	부품,소모품,자재,파츠	생활/건강>문구/사무용품>문구용품>화	0.8129258155823
5	두문 마커블랙보드 700x500 발광보드	부품,소모품,자재,파츠	생활/건강>문구/사무용품>문구용품>화	0.8029233813286
6	봉지 손잡이 100매 비닐 봉투1호 중 흰색	손잡이,잡이,비닐봉투,비닐,봉	생활/건강>주방용품>주방잡화>쓰레기	0.7840956449509

Sometimes provide incorrect results.

Next Tasks To Do

Next Step

- Vector database performance selection based on response time and complex queries.
- Compare the performance of different embedding models and select the best one.

Thank You for Listening